Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 16)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the number of plant species per m^2 in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	Mean plant species per m^2	Percentage of area covered by scrub / bushes
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

0 1 . 1 Explain what is meant by a **plagioclimax** community and why chalk grassland is an example.

[2 marks]

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0 1 . 2 Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

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0 1 . 3 Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

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0 1 . 4 Explain how burning maintains heathland.

[2 marks]

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0 2 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 2 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 3 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 3 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 3 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 4 Table 1 shows the estimated population of blue whales and the number killed by whaling in selected years.

Table 1

Year	1900	1930	1960	1986	2020
Estimated population	240 000	100 000	5 000	2 000	15 000
Whales killed that year	200	30 000	1 200	0	0

0 4 . 1 Calculate the percentage decline in the blue whale population between 1900 and 1986.

[1 mark]

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Answer [ ] %

0 4 . 2 Explain why whale populations were so easily over-exploited.

[3 marks]

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0 4 . 3 Describe how the International Whaling Commission regulates whaling.

[4 marks]

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0 4 . 4 Explain why the ITTO has been less successful than the IWC in achieving sustainable exploitation.

[2 marks]

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0 5 Figure 1 shows a simplified Antarctic food web.

Figure 1

Trophic level	Organisms
Producers	Phytoplankton, ice algae
Primary consumers	Zooplankton, krill
Secondary consumers	Small fish, squid, baleen whales, crabeater seals, penguins
Higher consumers	Toothfish, albatross, leopard seal, killer whale

0 5 . 1 Explain why the sea around Antarctica is so productive despite the cold.

[3 marks]

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0 5 . 2 Use Figure 1 to explain why fishing for krill and for Patagonian toothfish threatens other species.

[3 marks]

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0 5 . 3 Explain the importance of Antarctica in controlling global natural cycles.

[3 marks]

0 5 . 4 Explain why the area of winter sea ice around Antarctica has increased in recent years even though air and sea are warming.

[1 mark]

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0 6 . 1 State the Brundtland Commission definition of sustainable development.

[1 mark]

0 6 . 2 Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

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0 6 . 3 Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

0 6 . 4 Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

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0 7 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 7 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 7 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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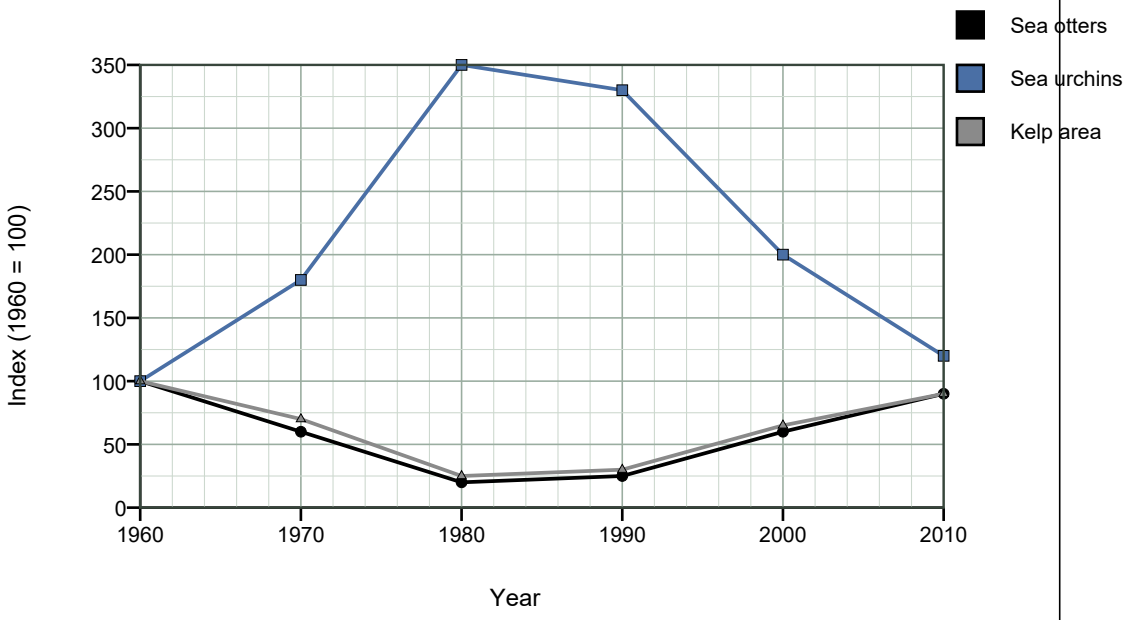
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0 8 Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

Figure 1



0 8 . 1 Describe the relationship between the three variables shown in Figure 1.

[3 marks]

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0 8 . 2 Explain the relationships shown in Figure 1.

[3 marks]

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0 8 . 3 Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

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0 8 . 4 Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

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0 9 Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

0 9 . 1 Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

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0 9 . 2 Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

0 9 . 3 Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

10

1 0 Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

1 0 . 1 Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

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1 0 . 2 Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

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1 0 . 3 Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

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1 0 . 4 Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS